

HapticServant – Delivering Haptic Products in a Virtual Reality Supermarket

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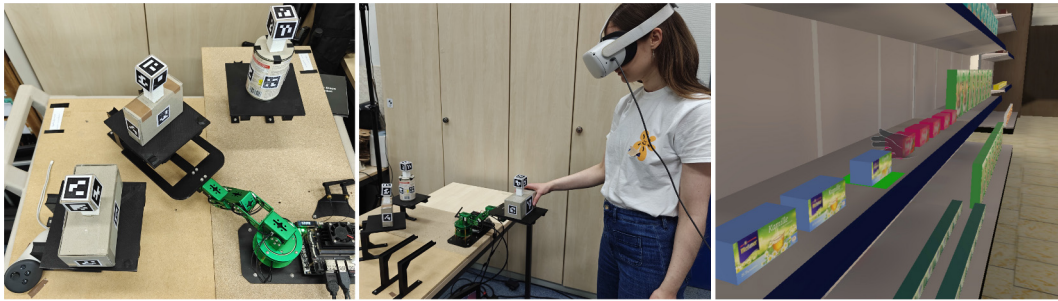


Figure 1: (Left) HapticServant, an encountered-type haptic system with three generic props which are tracked via ArUco markers; (Center) The robotic arm provides a generic prop that matches the shape of a grocery item; (Right) Simultaneously inside the virtual supermarket, the corresponding item is grabbed from the shelf to, e.g., examine its nutrition label.

ABSTRACT

This paper presents HapticServant, an encountered-type haptic system that utilizes a robotic arm to deliver passive props on demand, enhancing the shopping experience within a Virtual Reality (VR) supermarket simulation. The system comprises several 3D-printed components, including a modular palette mechanism that enables the presentation of tangible objects with various shapes and sizes. This enables users to physically grasp, rotate, and inspect items in a way that supports multisensory perception. In a user study ($N = 32$), a shopping task was evaluated across three different conditions: “Visual-Only Interaction” (without tangible props), “Single-Haptic Interaction” (one generic prop), and “Multi-Haptic Interaction” (three product-specific props). The evaluation focused on how different levels of robotic prop delivery affect presence, interaction quality, and task behavior. With props, participants described their experience as an improvement in terms of realism and engagement. The use of multiple, product-specific props maximized haptic fidelity for object handling, whereas a single generic prop may be sufficient when the task is focused on traversing the supermarket. HapticServant illustrates how simple and low-cost robotic encountered-type haptics can be leveraged for focused grocery item interactions, and can serve as a promising framework for similar VR applications beyond the supermarket context.

Index Terms: Virtual Reality, Haptics, Encountered-Type Haptic System, Immersion, Robot Arm, Virtual Supermarket, Passive Prop.

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1 INTRODUCTION

Virtual Reality (VR) enables the simulation of complex, controllable, and reproducible scenarios, with the potential for users to experience a strong sense of presence. This opens up possibilities to study human behavior or decision-making [8, 20, 22]. An example is virtual shopping, where tangible product interactions were, until now, a limiting factor for virtual-only supermarket simulations.

Haptic feedback is often omitted or reduced to controller vibration, which fails to replicate real-world tactile sensations. Yet, in tasks involving product interaction, such as grasping or rotating, the sense of touch plays a crucial role. Allowing users to inspect physical items from multiple angles supports reading nutrition labels or assessing shape and weight, enhancing immersion, which is linked to the sense of presence [12, 21]. A strong sense of presence is essential when the goal is to observe naturalistic behavior [24].

Robot-delivered haptics were proposed for general object manipulation [5, 18]. For our use case of tangible product interactions within a complex, interactive virtual supermarket, the robot arm (HapticServant) enables on-demand haptic feedback (an encountered-type system [28]). We aim for authentic consumer and behavioral research by allowing users to pick an item from the shelf, feel its weight and shape, and examine details such as the nutrition label. A robotic arm equipped with a custom-designed end effector (tool attached to the arm’s wrist) has been developed. The arm is connected to a Unity-based VR supermarket simulation and uses a hybrid tracking approach (HMD controller and ArUco markers [1]).

Our contributions are:

- The design and implementation of HapticServant, a low-cost, robot-driven system that delivers tangible props on demand in a VR supermarket (see Figure 1), demonstrating an accessible approach to encountered-type haptics.
- The development of three distinct experimental conditions for an evaluation, varying the level of tangible interaction:

– **Visual-Only Interaction (VOI):** The interaction is purely visual, without any haptic feedback.

- **Single-Haptic Interaction (SHI):** A single, generic physical prop (squared box) represents three differently-shaped virtual product types. An embedded VR controller tracks the position and rotation.
- **Multi-Haptic Interaction (MHI):** Three distinctly-shaped physical props, each corresponding to one product type. A webcam tracks multiple ArUco markers placed on the products.

With a study ($N = 32$), we want to answer the research questions:

- **RQ1:** What is the difference in perceived feeling of presence and interaction quality between VOI, SHI, and MHI?
- **RQ2:** How does the integration of robot-delivered haptic props (MHI) affect the system’s usability in a VR supermarket simulation compared to VOI?
- **RQ3:** How do SHI and MHI influence objective performance metrics such as task completion time compared to VOI?

2 RELATED WORK

A recent paradigm is on-demand tangible presentation. The robot’s role is not to simulate the object itself, but to act as a “haptic butler” that presents physical props to the user. The “Snake Charmer” pioneered this with its concept of interchangeable “endpoints” on a Robai arm, not only delivering objects with different shapes but also various textures and thermal properties [5]. Based on a UR5 arm, the “Alfred” system presents tangible, detachable objects. Once the user grasps the prop, they can manipulate it freely. The robot then intelligently repositions itself to provide haptic feedback for subsequent interactions [17]. This hybrid approach combines the realism of passive haptics with the flexibility of a robotic system.

The technical implementation varies for robotic hardware and tracking methods used (see Table 1). Stationary robotic arms are a common foundation. The “WYSIWYF” display utilized industrial robots, such as the PUMA 560, to achieve precise visual-haptic registrations [30]. Modern projects often utilize collaborative robots (cobots), such as the UR5, due to their safety features. These systems are typically limited to the arm’s static workspace, which led to developments such as the “per-plane reachability maps” in “H-Wall” to optimize interactions within this space [15]. Mobile robotic platforms address the workspace limitation by mounting the arm on a mobile base. The “CoboDeck” system exemplifies this architecture, combining a UR10 arm with a Robotnik RB-KAIROS omnidirectional platform. This introduces new challenges, such as real-time navigation and collision avoidance. Frequent tracking approaches utilize RGB-D cameras to track the user’s pose [5, 15]. Recent systems utilize HMD-based trackers, which are attached to the user’s hands, the robot’s end-effector, and tangible props [17].

2.1 Summary and Key Challenges

Studies consistently show that robotic haptic feedback significantly increases the sense of presence [14, 25]. The ability to feel material properties, as demonstrated by “ENTROPIA”, was found to be particularly convincing [18]. However, immersion can also be easily broken. Factors such as hearing the robot’s motors or experiencing noticeable delays while the robot repositions itself were found to be detrimental to the user experience [13]. These findings underscore the need for seamless, unobtrusive, and rapid robotic assistance.

Another primary challenge is latency. The robot must anticipate the user’s intent and move the physical prop into position before the user’s hand arrives. Failure to do so breaks the illusion. The evaluation of “Alfred” revealed that, even with optimized control schemes, the system required 2-3 seconds to position its prop, a delay that must be managed through interaction design [17]. Mobile systems such as “CoboDeck” address this by implementing a

Table 1: Projects in VR using robot arms for on-demand haptic feedback. The degree of complexity is judged on hardware cost, workspace size, software complexity, and tracking method.

	Haptic Type	Tracking Methods	Props	MB	Complexity
Araujo et al. [5]	hybrid	RGB-D	✓		low
Kim et al. [15]	active	RGB-D			middle
Mercado et al. [17]	hybrid	VR tracker	✓ (n.a.)	✓	middle
Mercado et al. [18]	active	VR tracker	✓		low
Mortezapoor et al. [19]	active	VR tracker, LiDAR, IMU, Odometry		✓	high
Vonach et al. [27]	active	optical			high
Yokokohji et al. [30]	active	optical			low
HapticServant	hybrid	optical, VR tracker	✓		low

n.a. = not automatic; MB = Mobile Base

“Chase” behavior, where the robot follows the user at a safe distance [19].

3 IMPLEMENTATION

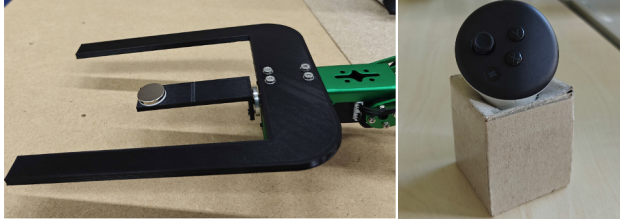
In the virtual supermarket environment, users can navigate by teleporting, and products can be picked from shelves and placed into a virtual shopping basket. A country-specific replica of a discounter (medium-sized supermarket) has been created [9–11]. It contains over 250 familiar products, and with a custom product creation pipeline, it is possible to mimic specific product shapes. Meta Quest 3 with Unity version 2021.3.3f1 and the XR Interaction Toolkit [3] is used.

3.1 Robotic Arm, End-Effector and Pallets Design

The robotic arm is positioned on a fixed table with haptic products placed around it. It enables the products to be delivered to the user only when needed. If no interaction is necessary, the user’s movement space can be kept free of physical elements. The arm can adjust the position of the haptic products depending on the height of the shelf on which the respective virtual product is placed.

The Yahboom DOFBOT [4] was chosen as the robot system for our project. This comprises a 6-DoF arm equipped with five 15 kg servo motors and one 6 kg servo motor. It possesses an arm span of 350 mm. The gripper can grasp objects with a diameter of up to 6 cm, and the payload capacity is a maximum of 500 g and 200 g when the arm is extended. The Jetson Nano 4GB serves as the robot arm’s computing unit. The ROS-TCP-Connector [26] was chosen as a software framework that manages communication between the robot arm and the VR system. This framework enables the bidirectional exchange of sensor data and control commands in real-time. A Python script was developed in ROS that defines all the required movement sequences of the robot. The script acts as a ROS node and subscribes to a topic where it receives movement commands as integer messages via the ROS-TCP-Connector. The robot’s basic logic is based on a finite-state machine.

Based on a 6 cm maximum grip size, the default mechanical gripper of the Yahboom DOFBOT is ill-suited for the supermarket setting. Furthermore, the gripper complicates intuitive human interaction by requiring precise object detection and release timing, and it can physically obstruct the user’s hand during the handover. A self-built end effector was designed, providing a key advantage in weight reduction at the end of the arm, thereby increasing the robot’s net payload. A forklift solution was developed with a 3D-printed fork mounted at the upper end of the arm (see Figure 2a), allowing the robot to pick up a pallet from below and transport it to the desired location, where the user then retrieves the object from it. Each product is placed on an exchangeable tray, from which it can be easily removed and returned. The trays are positioned on elevated stands. The props are equipped with magnets on their underside, which interact with a corresponding magnet on the end effector. The magnet on the end effector is strong enough to hold the product in place during transport and was designed to be rotatable.



(a) Fork attached to the arm with magnet on the end effector. (b) Prop with VR controller tracking for SHI.

Figure 2: Robot arm modification and generic haptic prop for SHI.

This allows the magnetic attraction to be deliberately weakened, enabling the object to be picked up without noticeable resistance.

3.2 Physical Props

Our approach is constrained by the robot arm's limited load capacity (<400 g), the pallet dimensions (15 × 15 cm), and the need for stability during movement. To represent the wide variety of items found in a supermarket with a manageable set of physical objects, we designed props based on generic shapes. This approach reflects the standardized forms common in real-world product logistics and allows a few props to stand in for numerous virtual items. The developed system supports three ArUco-based shapes (two boxes and one tin, see Figure 1 (Left)) and a single controller-based shape (see Figure 2b), each adapted in weight, surface texture, and stability.

In our case, three components need to be tracked: the user, the robotic arm, and, during product interaction, the passive prop. The Meta Quest 3 headset tracks the user's position and hands. For the two haptic conditions, it is also necessary to determine the robot's position. This allows the system to adjust the user's virtual location and rotation upon teleportation, ensuring that the virtual and real product positions align. To achieve this, one of the headset's VR controllers is placed at a fixed, predefined location next to the robot. From the controller's known pose, the system can determine the position and orientation of the robot's base, and by extension, the location of its pallet. For SHI, a script was implemented that is responsible for tracking the product with the controller inside. For MHI, product tracking relied on ArUco markers, which are attached to the passive props and can be detected by the webcam mounted on the headset. A Kalman filter (OpenCV library [2]) is used to reduce jitter and instability caused by image noise or partially occluded markers. This stabilizes the calculated position and orientation of the virtual product, preventing sudden jumps or erratic movement.

3.3 Visualization and Acoustic Shielding

A visual feedback mechanism was implemented for the virtual supermarket, indicating to the user which object provides haptic feedback and whether the passive prop is already available or the robotic arm has not yet reached its final position. If a scenario involves handing the product item back after inspecting it, visualization is necessary to indicate where the prop can be placed. Since the robotic arm cannot accept the prop at any arbitrary position within the shelf, the targeted placement ensures proper object handover.

Even when idle, the robot arm produces a noticeable noise due to the fan mounted on its CPU, and during rapid movements, the servo motors are noticeably louder. To suppress these potentially distracting sounds, participants wear Sony WH-1000XM5 noise-canceling headphones. A realistic supermarket soundscape, featuring the sounds of cash register beeps and the rattling of shopping carts, is played to mask the constant fan noise. To counteract the louder, intermittent motor sounds, the volume of this soundscape is dynamically increased for the duration of any robotic movement.

Table 2: Custom-designed questions aimed at the haptic experience.

Item	Question
HQ1	The physical object was correctly aligned with the virtual object.
HQ2	The product felt realistic in my hand.
HQ3	The weight of the object matched the visual representation.
HQ4	The shape of the object felt believable.
HQ5	The size of the object felt appropriate to its virtual counterpart.

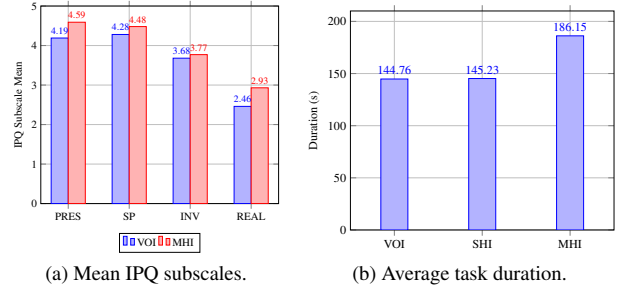


Figure 3: IPQ subscales [23] and task duration per condition.

4 EVALUATION

We aim to assess the potential of the three conditions, VOI, SHI, and MHI, to enhance the simulation of grocery shopping in VR.

4.1 Study Design

Demographics and prior VR experience were assessed at the beginning. Standardized questionnaires are utilized to measure the presence response with the Igroup Presence Questionnaire (IPQ) [23] and selected items from the Presence Questionnaire (PQ) [29], while usability is assessed through the System Usability Scale (SUS) [7]. A custom questionnaire with a 7-point Likert scale (1, 'fully disagree', to 7, 'fully agree') was used in the two haptic conditions to evaluate the realism of the physical props in terms of alignment, shape, weight, and size (see Table 2). User interactions based on task completion times, were logged for quantitative analysis.

The study was conducted in a lab environment. All participants were informed about cybersickness, their option to stop at any point, anonymous data collection, and, if they wished, they could revoke their participation (deletion of their data). Then they consented to the study and the use of their data. After a brief introduction phase with the Meta Quest 3's controls, each participant experienced the three different conditions. In each condition, participants performed an identical shopping task: they had to locate, pick up, examine, and return to the shelf three predefined products in the virtual supermarket. The order of the conditions was counter-balanced across participants to mitigate learning and fatigue effects. Following each one, they completed the questionnaires and participated in a brief, semi-structured interview.

4.2 Results

A total of 32 individuals participated with an average age of 24.19 years ($SD = 9.50$) and gender distribution of 53.1 % identifying themselves as male and 43.8 % as female. Additionally, 81.25 % of participants reported having little to no prior experience with VR.

4.2.1 Quantitative Measures

The normality of the data distributions was confirmed using the Shapiro-Wilk test. Subsequently, a paired-sample t-test was conducted. If the normal distribution was violated for the comparison of three conditions, the Friedman test was used. The significance

Table 3: Mean scores per item of the PQ [29].

Item	VOI	SHI	MHI
PQ3	2.53	3.81	3.50
PQ11	3.97	5.06	4.78
PQ12	3.34	5.25	4.88
PQ14	4.59	4.56	3.41
PQ23	3.06	4.28	4.38
PQ24	3.91	4.88	4.56

Table 4: Mean scores of the custom haptic experience questions.

Item	SHI	MHI
HQ1	3.84	3.63
HQ2	3.38	5.22
HQ3	3.19	4.88
HQ4	2.69	5.63
HQ5	3.41	5.25

level was set to be $p < 0.05$. The practical relevance of the findings was assessed by using Cohen’s d . For the PQ and HQ questions, Cronbach’s alpha was used to measure the internal consistency, with $\alpha > 0.7$ being acceptable.

The IPQ’s PRES subscale (see Figure 3a) was significantly higher in the MHI condition compared to the VOI condition ($t(31) = -2.75, p = 0.01, d = 0.49$). The results from the PQ are detailed in Table 3 ($\alpha > 0.7$). Comparing the VOI and MHI conditions for PQ, significantly higher ratings were observed in the MHI condition for all items except PQ14 (targeting the delay, which, with roughly 3 seconds, was highest for MHI). PQ14 was therefore significantly lower compared to VOI ($\chi^2(2) = 7.33, p = 0.03$). SHI outperformed MHI slightly for most items (except PQ23). However, these differences were not statistically significant.

The calculated SUS score was **75.55** “Good” [6] in the VOI condition and **75.86** “Good” [6] in the MHI condition.

As shown in Table 4 ($\alpha > 0.7$), the average rating regarding the alignment accuracy between the physical object and virtual counterpart (HQ1) was slightly higher in the SHI condition, though this difference was not significant ($t(31) = 0.53, p = 0.60, d = 0.003$). In contrast, mean ratings for items HQ2 to HQ5 were significantly higher in the MHI condition ($t(31) < -4.96, p < 0.001, d > 0.88$).

As shown in Figure 3b, participants in the MHI condition required significantly more time to complete the tasks than in the other two conditions ($\chi^2(2) = 23.25, p < 0.0001$).

4.2.2 Qualitative Feedback

Based on the interview, for VOI, participants criticized the need to twist their arm or transfer the product to the other hand when they wanted to view the nutrition label. Also, in this condition, some participants unintentionally dropped the virtual products on the ground. These difficulties did not occur in the two haptic conditions. Objects could be held more securely and rotated intuitively, even with just one hand. Although the system’s capabilities had been explained beforehand, several participants did not realize during the trial that the robotic arm was handing them the props. In the interviews, three participants reported they had assumed they were picking up and returning the objects at their respective product stations. The haptic experience in the MHI condition was particularly described as highly convincing. A clear preference for haptic feedback emerged, with no participant preferring the VOI condition. Although in the two haptic conditions, some participants reported brief disorientation when teleported in front of a haptic product. An overwhelming majority (30 out of 32) favored the MHI condition, while the remaining two preferred the SHI condition.

4.3 Discussion

The evaluation confirmed that enabling users to physically grasp and manipulate products via HapticServant significantly increased their sense of presence (significantly higher IPQ scores and PQ ratings for the MHI and SHI conditions, respectively). This finding suggests that the ability to perform natural actions, such as feeling an object’s weight or turning it to inspect a label, is a key driver of presence in the context of a virtual grocery shopping simulation. This result is supported by qualitative data, based on interview

statements (favoring MHI). Regarding **RQ1**, a clear difference in the sense of presence and interaction quality emerged between the two haptic conditions. Both haptic setups increased presence over VOI. The MHI condition achieved an “Excellent” [16] for the subscale PRES. For the PQ items, SHI was slightly favored. This trend was reversed by the custom questions and qualitative feedback. The haptic conditions produced a seamless experience, further enhanced by effective acoustic shielding, which prevented most users from being consciously aware of the robot’s movements and noise. Furthermore, when comparing the two haptic conditions directly, participants strongly preferred MHI based on product-specific props with distinct shapes and weights, even though the arm took slightly longer to respond. The custom questions (HQ2-HQ5) confirm experience ratings for shape, weight, and size in MHI.

In response to **RQ2**, the integration of robot-delivered props did not negatively affect the system’s usability. The SUS scores showed no statistically significant difference between MHI and VOI. This finding is important, as it indicates that the added hardware complexity did not compromise the perceived usability. This quantitative result is reinforced by qualitative feedback, as participants rarely reported frustration or increased mental effort in the haptic conditions, despite the additional system components and occasional tracking delays. In addition, noise was successfully shielded through the use of noise cancellation in combination with the dynamically adjusted supermarket soundscape.

Regarding objective performance (**RQ3**), the introduction of physical props revealed a clear trade-off. Task completion time was significantly longer in the MHI condition, where, besides an increase (roughly 3 seconds) for the robot arm movement, users’ tendency was to spend more time exploring the tangible objects. However, this slower pace was offset by a reduction in interaction errors; participants in the haptic conditions dropped items less frequently. Qualitative observations support this interpretation, as the visual-only condition was often described as faster but less precise, particularly when rotating or stabilizing products.

In summary, the results validate that the HapticServant approach successfully achieved its goal of increasing the presence response in the VR environment without degrading usability. The study highlights that users are willing to accept a moderate decrease in task efficiency in exchange for a significantly more realistic and engaging experience. Both quantitative and qualitative data confirm that, from the user’s perspective, the benefits of a differentiated MHI outweigh the current limitations of the system. The results suggest that the benefit of multiple, product-specific props is task-dependent: while increased haptic fidelity supports detailed product perception (MHI), tasks focusing on general navigation or item localization in the supermarket do not necessarily require such a level of physical interaction (SHI is sufficient, backed by the slightly higher PQ ratings).

5 LIMITATIONS AND FUTURE WORK

One of the limitations of the current prototype is the small range, limited strength, and restricted flexibility of the robotic arm used. The setup is also restricted to a single tangible product handed over at a given time. This limitation also requires a visual indicator to show which product is currently tangible, thereby reducing the overall realism of the simulation. These constraints could be addressed with the help of a larger robot arm. A robot with increased reach could additionally provide access to more than three different shapes. Additionally, when combined with an eye-tracking system, the system could identify which product a user is attempting to reach for and automatically move the corresponding prop into position. This can mitigate some of the delay through the robot arm. Another possible improvement would be placing several identical robot systems around the user, as suggested in project “VRRobot” [27]. If there were four robot arms, each robot arm

would only need to handle products from shelves with the same orientation. This would eliminate the need to rotate the virtual world during the simulation, thereby reducing short-term disorientation.

6 CONCLUSION

Immersion in a virtual supermarket simulation was enhanced through the use of affordable hardware, allowing users to grasp, feel, and inspect tangible products that were delivered on demand by a robotic arm. The sense of presence increased with the use of physical props without compromising usability. Comparison between haptic conditions (SHI and MHI) revealed task-dependency: increased haptic fidelity (MHI) is helpful when the aim is to differentiate shape, weight, size, or even texture of products, while tasks focusing on, e.g., the layout of the supermarket do not necessarily require such a level of physical interaction (SHI is sufficient).

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